Solution

Here the number of unknowns is 3.

The matrix form of the system is $AX = B$, where $A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 1 & \lambda \\ 1 & 3 & -5 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 7 \\ \mu \\ 5 \end{bmatrix}$.

Applying elementary row operations on the augmented matrix $[A \mid B]$, we get

$$[A \mid B] = \left[\begin{array}{ccc|c} 1 & 2 & 1 & 7 \\ 1 & 1 & \lambda & \mu \\ 1 & 3 & -5 & 5 \end{array}\right] \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 7 \\ 1 & 3 & -5 & 5 \\ 1 & 1 & \lambda & \mu \end{array}\right]$$

$$\xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - R_1,} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 7 \\ 0 & 1 & -6 & -2 \\ 0 & -1 & \lambda - 1 & \mu - 7 \end{array}\right] \xrightarrow{R_3 \to R_3 + R_2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 7 \\ 0 & 1 & -6 & -2 \\ 0 & 0 & \lambda - 7 & \mu - 9 \end{array}\right].$$

(i) If $\lambda = 7$ and $\mu \neq 9$, then $\rho(A) = 2$ and $\rho([A \mid B]) = 3$. So $\rho(A) \neq \rho([A \mid B])$. Hence the given system is inconsistent and has no solution.

(ii) If $\lambda \neq 7$ and μ is any real number, then $\rho(A) = 3$ and $\rho([A \mid B]) = 3$.
So $\rho(A) = \rho([A \mid B]) = 3 =$ Number of unknowns. Hence the given system is consistent and has a unique solution.

(iii) If $\lambda = 7$ and $\mu = 9$, then $\rho(A) = 2$ and $\rho([A \mid B]) = 2$.

So, $\rho(A) = \rho([A \mid B]) = 2 <$ Number of unknowns. Hence the given system is consistent and has infinite number of solutions. ■

EXERCISE 1.6

1. Test for consistency and if possible, solve the following systems of equations by rank method.
   (i) $x - y + 2z = 2, \quad 2x + y + 4z = 7, \quad 4x - y + z = 4$
   (ii) $3x + y + z = 2, \quad x - 3y + 2z = 1, \quad 7x - y + 4z = 5$
   (iii) $2x + 2y + z = 5, \quad x - y + z = 1, \quad 3x + y + 2z = 4$
   (iv) $2x - y + z = 2, \quad 6x - 3y + 3z = 6, \quad 4x - 2y + 2z = 4$

2. Find the value of k for which the equations $kx - 2y + z = 1, \quad x - 2ky + z = -2, \quad x - 2y + kz = 1$ have
   (i) no solution (ii) unique solution (iii) infinitely many solution

3. Investigate the values of λ and μ the system of linear equations $2x + 3y + 5z = 9$, $7x + 3y - 5z = 8$, $2x + 3y + \lambda z = \mu$, have
   (i) no solution (ii) a unique solution (iii) an infinite number of solutions.

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